

David Khachatryan - Week 7 Report

Summary

This week our focus was achieving fully controlled flight. We were not happy with the PID tuning we had done before. During the class we redid our PID tuning by following the Ziegler–Nichols method. First, we increased the I gains as much as possible until the drone was quickly oscillating. Then, we added damping to the point where the oscillations were reduced to a steady, sustained level without growing or decaying. Upon restarting the flight, we further tuned our I gains. It rose from 0.1 to 0.7, which reduced drifting to the point that we successfully moved the drone to an identified position and held it idle.

What went well

Retuning our parameters made the flight way more controllable. We still noticed a little bit of drift, but upon increasing the I gains, the drift was manageable. The Ziegler–Nichols method helped us achieve controlled flight.

What did not go well, why?

We spent way too much time tuning our amplitudes. In this process, we went into the wrong direction of increasing our amplitudes for faster response. This felt like going from driving an automatic car to a stickshift. With faster input response, the drone went out of control more frequently. Initially, we thought this was due to higher neutral thrust when the battery was fully charged. This took us about an hour to eventually go to 5 degrees of roll and pitch amplitude.

What will you change for next class

For the next class, I would need to focus more on yaw control. We could potentially lower P gains for yaw even more. For the purpose of controlled flight, I had to do a lot of manual yaw input and I was not very good at it. I certainly need to work on manual control and tune it further if necessary.

Contributions

We discussed the procedure for the Ziegler–Nichols method and implemented it. Both of us were on top of it, so our contributions were 50/50.